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Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled pH conditions

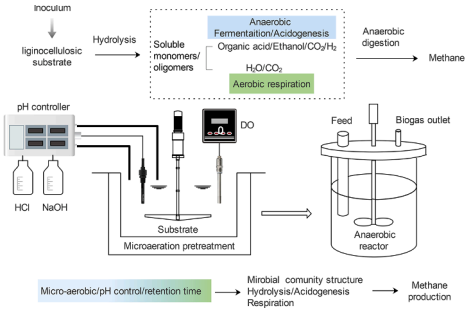
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HIGHLIGHTS

- Proteiniphilum and Rummeliibacillus were dominant during mirobial pretreatment.
- Pretreatment enhanced hydrolysis and acidogenesis of lignocellulose.
- Both pH and pretreatment time greatly affected methane yield.
- The highest methane yield was found at pH 8 and 12 h of pretreatment.

GRAPHICAL ABSTRACT



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ABSTRACT

The effects of various microaeration strategies and process parameters on anaerobic digestion (AD) of lignocellulosic substrates have received increased attention; however, different results have been reported. To determine optimal conditions and clarify the mechanisms influencing this process, the effect of pretreatment of microaerobic microbial on corn stover decomposition and AD was investigated with real-time pH control. Fresh cow manure was chosen as the inoculum, as it has the strongest cellulose hydrolysis capacity under microaeration conditions. Microaeration microbial pretreatment effectively promoted the hydrolysis and acidogenesis of corn stover, and pH considerably affected total solid reduction, volatile fatty acid (VFA), and accumulation of soluble chemical oxygen demand (sCOD) patterns by shifting microbial communities. Different pH levels and pretreatment times led to positive and negative effects on methane yield. A 12-h pretreatment of substrate at pH 8 prior to AD increased the methane yield by 16.6% in comparison with the un-pretreated sample.

1. Introduction

China has strived to vigorously develop the anaerobic digestion (AD) of agricultural wastes to produce bio-natural gas since 2015 for environmental protection, renewable energy alternatives, and CO_2 emission reduction. Corn stover, accounting for about 32.5% of agricultural straw produced in China, was regarded as a good candidate. Many biogas and bio-natural gas projects using corn stover as the main

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substrate have been built and operated in China in recent years, but the project profit depends on government subsidies has limited rapid development. The low degradation efficiency due to the recalcitrant structure of lignocellulose is one of the important limiting factors (Woiciechowski et al., 2020).

It is well known that pretreatment is an effective measure for improving digestion rate, thereby enhancing methane generation from lignocellulosic substrates. Many pretreatment technologies, such as alkaline, acid and steam explosion, are investigated for improving the lignocellulosic biomass AD (Frigon et al., 2012), with good results achieved. However, these pretreatment technologies usually have specific drawbacks, such as increased energy consumption, inhibitor formation, or scaling-up difficulties. As a result, few of them have been used in commercial production, because of negative economic profit or the introduction of pollution.

Accordingly, enhanced microbial hydrolysis pretreatment under microaeration conditions is considered an effective method to improve the AD of lignocellulosic substances by promoting certain facultative and aerobic microbial activities, while this increases biomass dissolution (Botheju & Bakke, 2011). However, the effect of microaeration on the AD of lignocellulose substances remains controversial. It is extensively reported that microaeration has positive effect (Fu et al., 2016; Lim & Wang, 2013), but some studies found that the addition of oxygen deteriorated AD, leading to reduced methane yield (Johansen & Bakke, 2006). The uncertain outcomes may arise from different microaeration strategies, techniques, or intensities.

As reported in previous studies, microaeration improves the methane potential of biomass by effectively promoting hydrolysis and acidogenesis of complex substrates into simple organic matter (Nguyen et al., 2019). Hydrolysis, which is positive for AD, and consumption of substrate, which is negative for AD, exist in the pretreatment process. It can be speculated that if oxygen is supplied improperly, or if the pretreatment time is too long, then oxygen respiration is too high and the availability of substrates for AD is decreased too much, leading to an adverse impact on AD. To maximize methane yield, pretreatment should be controlled to maximize the hydrolysis of lignocellulose while simultaneously minimizing chemical oxygen demand losses.

There are two kinds of microaeration strategies used for enhancing the AD of biomass (Chen et al., 2020): one is to dose a little amount of oxygen (dissolved oxygen DO, 0.1–1.0 mg/L) or air to an anaerobic bioreactor (Nguyen & Khanal, 2018), and the other is to carry out the microaeration process as a pretreatment before AD in a separate reactor. The latter strategy has been adopted in many biogas projects using lignocellulosic substrates as feedstock. However, the effects of microaerobic pretreatment and pretreatment process parameters on the AD performance of lignocellulosic substrates and the influencing mechanisms of the process are less reported. The pH and length of pretreatment are important controllable parameters for pretreatment outcomes. The present work examined how pH and the length of microaerobic microbial pretreatment (MMP) affected the corn stover AD. Besides, inoculum type, total solid (TS) loss, organic acid production, cellulase activity, and microbial community structure were also studied to reveal the mechanism of positive and negative effects of MMP on the AD performance of corn stover.

2. Materials and methods

2.1. Raw material and inoculum

This study obtained the dry corn stover in the Licheng District, Jinan, China. Fresh cow manure, which contained 131.6 g/L TS, was collected from Jinan Jiabao Dairy Co., Ltd., and stored at 4 °C for no more than 7 d before use. We obtained the anaerobic sludge containing 38.09 g/L volatile solids (VS) and 63.26 g/L TS in the mesophilic digestion tank of the upflow anaerobic sludge blanket in Shandong Xingguang Sugar Group Co., Ltd., Dezhou, China.

2.2. Experimental methods

2.2.1. MMP process

Open reactors with water jackets and mechanical mixers were used to conduct the MMP. Corn stover and inoculum were mixed at a TS ratio of 3:1, followed by the addition of water until the TS content was 7.5%. The temperature in all reactors was maintained at 35 °C. Initially, the reactors were then subjected to 5 min of N_2 flushing for achieving soluble air replacement. Subsequently, the DO concentration was kept in the range of 0.1–1 mg/L. This concentration range was found to provide microaeration conditions for microaerophilic and facultative microbes (Wu et al., 2015). Such conditions may be achieved by mixing in open air. An online DO detector was used to monitor the DO concentration in the broth. The pH and pretreatment time were the examined variables at different levels in this study. The examined pH range of the pretreatment experiment was 5 to 8, which can be formed through process control or can take place spontaneously. The pH was controlled in real-time with a pH controller composed of a pH sensor, pH control unit, and charging pump; 1 M HCl or 1 M NaOH was added when the pH deviated from set limits. Each test was performed in triplicate. Pretreatment slurry samples collected for testing were stored at −20 °C before analysis.

2.2.2. TS reduction test of corn stover during MMP

We used a cutting mill to cut corn stover into small pieces, and then a 10–20 mesh was used for filtering. To reduce the experimental error, the TS reduction of corn stover in the MMP was investigated by enclosing the substrates in nylon bags (pore size, <100 μm). (This step was an innovation of the current work). Then, approximately 1.0 g dry substrate was added to the nylon bag, the bag was then sealed thermally, and added to the MMP reactor. The substrate sealed in the bag was able to contact microaerobic microbes. Each nylon bag was obtained following pretreatment and rinsed by running water, following by drying within the oven under 105 °C. The TS of the substrate before and after pretreatment was weighed to calculate TS reduction.

2.2.3. Experimental setup for AD

The automatic methane potential test system (Bioprocess Control, Lund, Sweden) was used to determine the methane production of substrates. This system can determine methane generation of diverse samples as time goes by. For diverse samples, their AD was conducted within the 500 mL bottles in batches. The anaerobic sludge was mixed with substrates at 3:1 (VS) into the bottles (Li et al., 2013). Thereafter, the resultant mixtures were subjected to incubation under 38 °C. Additionally, we selected another bottle that contained inoculum, without substrate to operate at the same conditions as control. Each blank and tests were conducted thrice. Finally, the endogenous control production was subtracted to obtain the net methane production.

2.3. Analytical methods

The approaches to examine wastewater and water were used to determine VS and TS (APHA, 2005). To determine soluble chemical oxygen demand (sCOD), we collected the sample supernatants obtained after 10 min of centrifugation at 12 000 rpm. sCOD was determined using a 5B-3B (V8) water quality detector (Beijing Lianhua Technology Co., Ltd., China).

Then, the fiber analyzer (ANKOM 220, USA) was employed to determine lignin, hemicellulose, cellulose and the soluble substances in corn stover by Van Soest et al.'s method (1991). We also determined the amounts of neutral detergent fiber (NDF), acid detergent lignin (ADL) and acid detergent fiber (ADF) within dry samples. To be specific, soluble substance levels were determined to be the difference of TS with NDF, while hemicellulose level was measured to be the difference of ADF with NDF, the cellulose level was measured to be the difference of ADF with ADL, whereas lignin equaled to ADL.

The Agilent 7890 series gas chromatography system was adopted to

determine the concentrations of volatile fatty acids (VFAs; acetic, propionic, butyric, isobutyric, valeric, and isovaleric acids). Prior to VFA measuring, sample pretreatment was completed. After centrifugation, samples were subjected to acidification using the 3% phosphoric acid until the pH value was below 2, for the sake of converting fatty acids to the corresponding undissociated states. Thereafter, deionized water was used to dilute samples so that the VFA contents were within the standard curve range, meanwhile, the 0.22-μm filters were used for filtering. In this experiment, we used the HP-FFAP column (dimension, 50 m × 320 μm × 0.5 μm). Additionally, we adopted the flame ionization detector, with the operating temperature being set at 300 °C. In addition, the nitrogen carrier gas was injected constantly at 30 mL/min, with an inlet temperature being 250 °C. The original temperature of the oven was 60 °C, which was then heated at a 10 °C /min heating rate to 100 °C. The temperature at 100 °C was held for 2 min, then the temperature of the oven slowly elevated at 10 °C /min to 250 °C.

2.4. Extracellular carboxymethyl cellulase assays

The pretreatment slurry was subjected to 10 min of centrifugation at 12 000 rpm and 4 °C to obtain the supernatants for enzyme assays. Ghose (1987) was adopted to assay the carboxymethyl cellulase (CMCase). Later, 1 mL of the 0.05 M pre-heated acetate buffer (pH 6.0) that contained carboxymethyl cellulose (0.5%, w/v) was added into the 0.5 mL supernatants. Afterwards, the resultant mixture was subjected to 1 h of incubation under 50 °C. Besides, Miller's dinitrosalicylic acid approach Miller (1959) was utilized to assay the production of reducing sugars within the enzyme-exposed samples. Under the experimental conditions, one enzyme unit referred to the enzyme volume releasing 1 mmol/min reducing sugars.

2.5. Kinetic study

The methane yield potential and the hydrolysis constant during AD was estimated using the first-order kinetic equation, which was written as:

$$B_t = B_0(1 - e^{-kt}) \quad (1)$$

where B_t stands for the accumulated methane amount generated at time t (d) (unit, L CH_4/g TS), B_0 represents the potential of methane production (L CH_4/g TS), while k indicates the constant of hydrolysis kinetics (d^{-1}). The Curve Expert (version 1.4) was used to adjust the paired experimental data (B, t) through non-linear regression, which made it possible to calculate k, R^2 and B_0 at 95% confidence interval (CI) for every case investigated.

2.6. Microbiological analysis

High-throughput sequencing was used to analyze the microbial community structure of pretreatment slurry before and after five days of MMP. We used the centrimonium bromide approach to extract the total genomic DNA. Later, we applied specific primers (including 16S V4: 515F- 806R, 18S V4: 528F-706R, and 18S V9: 1380F-1510R) that had barcodes to amplify diverse regions in 16S rRNA/18SrRNA/ITS genes (16S V4/16S V3/16S V3-V4/16S V4-V5, Arc V4, ITS1/ITS2 and 18S V4/18S V9). Each PCR procedure was conducted using the Phusion® High-Fidelity PCR Master Mix (15 μL, New England Biolabs, USA). Thereafter, the Qiagen Gel Extraction Kit (Qiagen, Germany) was employed to purify PCR products.

We adopted the TruSeq ® DNA PCR-Free Sample Preparation Kit (Illumina, USA) to generate sequencing libraries in accordance with specific protocols, followed by the addition of index codes. Besides, the Agilent Bioanalyzer 2100 system and the Qubit@ 2.0 Fluorometer (Thermo Scientific, USA) were employed to assess the quality of libraries. At last, the Illumina NovaSeq platform was adopted for sequencing the libraries, which produced the paired-end reads (250 bp in length). In addition, Uparse (Uparse v7.0.1001, http://drive5.com/uparse/) was applied in sequence analysis (Edgar, 2013). At last, we deemed sequences that showed ≥ 97% similarity as the identical operational taxonomic units (OTUs).

2.7. Calculation methods

SPSS22.0 (SPSS Inc., Chicago, USA) was adopted for data analysis. All results were expressed as mean ± SD. Statistical significance was determined by analysis of variance (ANOVA). We deemed the significance level as $p < 0.05$. In the presence of statistical significance, the means were separated using Tukey's HSD test.

3. Results and discussion

3.1. Effect of inoculum type on MMP

The MMP efficiency is mostly determined by certain facultative hydrolytic microbes that have the ability to degrade lignocellulose, it is important to have as many microbes as possible in the inoculum (Wen et al., 2015); the choice of inoculum for pretreatment is important. Composing microbial agent, cow manure, aerobic sludge, and anaerobic sludge were tested for their cellulose hydrolysis ability under MMP in this study.

The filter paper weight loss method was adopted to record the cellulose hydrolysis activity. Ashless filter paper packed in a nylon bag was added to the reactor during MMP with different inocula for 2 d. The dry weight of the paper was weighed before and after pretreatment. The results showed that the usage of composing microbial agent, aerobic sludge, and anaerobic sludge as inocula did not lead to significant TS reduction, whereas the fresh cow manure inoculation test led to prominent filter paper TS loss (10.8%). Hence, it was concluded that the microbial community in fresh cow manure can effectively hydrolyze lignocellulose under microaeration conditions. Fresh cow manure was used as the inoculum in the next MMP experiment. The results of the aerobic and anaerobic sludge inocula are consistent with the reports of Yu et al. (2017) and Tsapekos et al. (2017).

3.2. Hydrolysis and acidogenesis of corn stover at different pH MMP

3.2.1. TS reduction of corn stover during MMP

As reported in previous studies, hydrolysis accounts for a rate-limiting link during the organic matter AD process (Song et al., 2019), especially lignocellulosic material. In this study, all the microaeration pretreatment experiments maintained a DO concentration between 0.1 and 1.0. g/L. This DO concentration may enhance solubilization of the organic matter by promoting the activity of microaerophilic and facultative microorganisms (Botheju & Bakke, 2011). The TS reductions of corn stover, sCOD accumulation, total VFA concentration, NH_4-N concentration, and extracellular CMCase activity were measured at different processing times (1 d, 2 d, 3 d, 4 d, 5 d) under different pH control strategies (pH 5, pH 6, pH 7, pH 8, and an uncontrolled pH (pH-U)) (Fig. 1). MMP effectively promoted hydrolysis of corn stover (Fig. 1a). Comparing different pH control strategies, it can be seen that TS reduction was greatly affected by pH during MMP. The hydrolysis sequence of the samples was pH 8 > pH 7 > pH-U > pH 6 > pH 5. This result is consistent with Zhang et al. (2012), and differs from that of Chen et al. (2007), whose experimental results showed that the hydrolysis of lignocellulose-type sludge at different pH values was in the following order: alkaline > acidic > neutral/blank test. After 5 d of pretreatment, the highest TS reduction reached 16.8% at pH 8, which was 33.9% higher than that of the pH-U sample. The lowest TS reduction reached 9.7% at pH 5, which was 30.6% lower than that of the pH-U sample. The slow rate of hydrolysis under acidic conditions may be due to the inhibition of microbial hydrolysis, as mentioned by Van

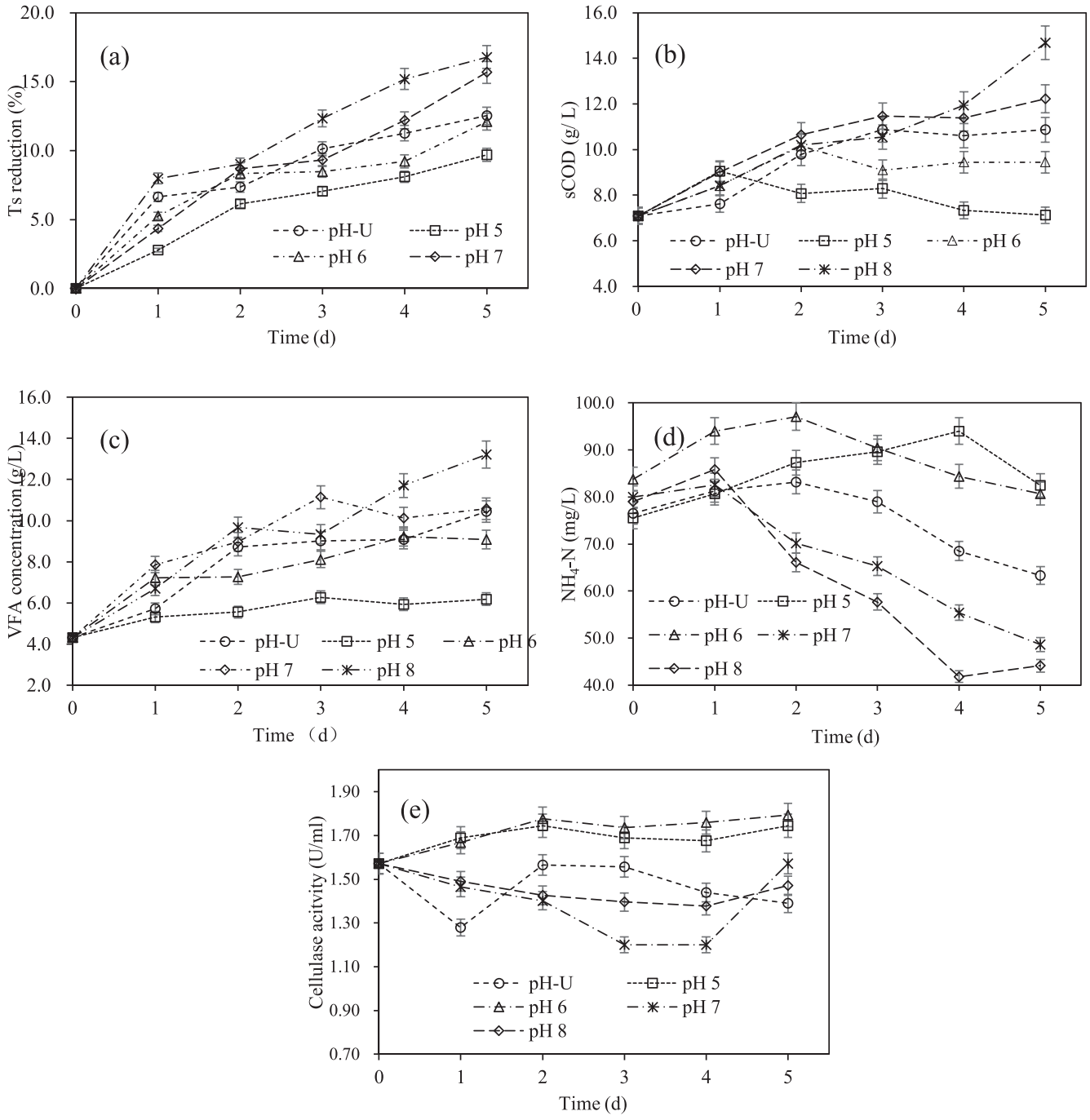


Fig. 1. Kinetic analysis of TS reduction of corn stover (a), sCOD accumulation (b), total VFAs concentration (c), NH_4-N concentration (d) and extracellular CMCase activity (e) determination during MMP.

Ginkel and Logan (2005), who discovered that the suppression effect taking place at pH = 5 was associated with the pH value because of the impact on undissociated acid content.

To clarify the specific roles and mechanisms of MMP in lignocellulose hydrolysis, the components (soluble components, cellulose, hemicellulose, and lignin) were measured pre- and post-hydrolysis under

Table 1
The component contents of corn stover before and after 5 days of MMP.

Sample	Lignocellulose/Substrate(%)	Cellulose/Lignocellulose(%)	Hemicellulose/Lignocellulose(%)	Lignin/Lignocellulose(%)
Raw material	88.42 ± 0.45	57.97 ± 0.32	28.73 ± 0.26	13.30 ± 0.15
pH 5	93.74 ± 0.52	57.12 ± 0.30	28.18 ± 0.27	14.70 ± 0.11
pH 6	94.53 ± 0.54	57.22 ± 0.38	27.85 ± 0.26	14.93 ± 0.13
pH 7	95.72 ± 0.48	56.84 ± 0.29	27.44 ± 0.27	15.71 ± 0.14
pH 8	96.58 ± 0.62	56.89 ± 0.40	26.72 ± 0.21	16.39 ± 0.14
pH-U	94.37 ± 0.44	57.50 ± 0.34	27.31 ± 0.24	15.18 ± 0.12

diverse pH values to calculate changes in composition (Table 1). The results indicated that soluble components accounted for a large proportion of TS reduction during pretreatment, especially on the first day. The proportion of lignin increased after MMP, because the degradation of cellulose and hemicellulose was faster than that of lignin. There was no obvious difference between the degradation rates of cellulose and hemicellulose, differing from Fu et al. (2015), who found that hemicellulose and lignin were degraded faster than cellulose by microaeration-enhanced AD of corn straw.

3.2.2. sCOD accumulation and organic acid production during MMP

MMP elevates the sCOD content and promotes solubilization, which plays an important role in promoting methane generation. The sCOD of hydrolysate was greatly affected by pH (Fig. 1b). At pH 5, after an increase in the first day, the sCOD concentration declined over time. The sCOD concentration at pH 6 began to decline slowly after 2 d, while at pH 7 and pH 8, the sCOD concentrations increased over time.

After 5 d of pretreatment, the order of sCOD concentration in the hydrolysate was consistent with the TS reduction results. This result is due to the faster hydrolysis of corn stover at higher pH values. Wang et al. (2016) found that neutral pH was more favorable for sCOD accumulation. Comparing TS and sCOD accumulation during pretreatment, at pH 5 and pH 6, corn stover was hydrolyzed every day during the 5 d pretreatment, but the sCOD concentration in the hydrolysate rose and fell. This means that the speed of sCOD consumption was faster than sCOD generation, which negatively impacts methane production. Even at pH 7 and pH 8, the velocity and magnitude of sCOD increase were lower than TS reduction, indicating that sCOD was massively produced and consumed simultaneously. Considering these results, for maximum methane production, the pretreatment time should be no more than 2 d.

VFAs are the most important products of hydrolysis and acidogenesis of corn stover during MMP. As the precursor of anaerobic methanation, VFA concentration and composition have a great impact on AD (Li et al., 2020a). To be specific, total VFAs are calculated by adding the propionic, acetic, isobutyric, butyric, isovaleric and valeric acid contents. Fig. 1c displays the mechanism of pH in VFAs composition and generation in corn stover pretreatment. The VFA accumulation trends are similar to those of the sCOD concentration of hydrolysate. A higher pH was beneficial to VFA accumulation. After 5 d of pretreatment, the concentrations of VFAs were 6.18, 9.09, 10.58, 13.22, and 10.44 g/L at pH 5, pH 6, pH 7, pH 8, and pH-U, respectively. Such findings were consistent with those obtained by Karthikeyan et al. (2016), where the VFA content within the traditional continuous stirred tank reactor increased by 35% at pH = 9 relative to that at pH = 6 in the processes of kitchen waste hydrolysis and acidification.

The VFA:sCOD ratio is an indicator of acidogenesis. It was observed that the VFA:sCOD ratio in the hydrolysate gradually increased with the length of pretreatment. Initially, the proportion of VFAs equivalent to sCOD in total sCOD was 61.2%, but after 5 d of pretreatment, the proportion increased to 87.6%, 96.2%, 87.5%, 90.4%, and 96.3% at pH 5, pH 6, pH 7, pH 8, and pH-U, respectively. This means acidogenesis was faster than hydrolysis, and that hydrolysis was the rate-limiting process. In addition, on the first day, the sCOD of the pH 5 pretreatment was the highest, whereas the amount of VFAs was the lowest, which means that lower pH had an inhibitory effect on acidogenic fermentation.

In a previous study, pH was found to greatly influence the type of acidogenic fermentation (Li et al., 2020b). The present work indicated that pH significantly affected the VFA composition (Table 2). In all pretreatment samples, acetic acid had the largest proportion, accounting for 77%–86% of VFAs, followed by propionic acid. This result was similar to that of Wang et al. (2016), who found that the level of acetic and butyric acid, which accounted for 76.75% of total VFAs, was optimized after the kitchen waste hydrolysis and acidification processes. The high acetic acid content is beneficial to AD as acetic acid, a precursor for methane production, contributes to 65%–95% of methane production (Weber et al., 1984). The proportion of propionic acid was most affected by pH control. A low pH value predicts the low propionic acid percentage, which may be due to a metabolic pathway shift at different pH values (Sawatdeenarunat et al., 2015). It was reported that propionic acid accumulation can inhibit methanogenesis, and that the promoted formation of acetate and the reduced accumulation of propionate were good for methane generation (Wang et al., 2006). The acetic acid:propionic acid ratio in the hydrolysate is regarded as an indicator of the stability of AD; a ratio above 2.0 is considered safe for AD (Wagner et al., 2014). In this study, the propionic acid concentration and the acetic acid:propionic acid ratio were far from inhibitory levels, and this was to the advantage of the later utilization of VFA during methanogenesis.

3.2.3. CMCase activity during pretreatment

Extracellular cellulase activities were examined during MMP (Fig. 1e). The results showed that the inoculum had high CMCase activity, and that all the test samples maintained high CMCase activity during pretreatment. The extracellular CMCase activities of pH 6 and pH 5 hydrolysate were higher than those of the other pH values, but the higher extracellular CMCase activities did not lead to higher TS reduction, which means that extracellular CMCase activities cannot represent the actual cellulose hydrolysis ability in the MMP process. This is because, unlike cellulose-producing fungi, few bacteria can excrete extracellular cellulase. Most cellulases produced by bacteria are intracellular enzymes, which are combined with the cytomembrane (Sadhu & Maiti, 2013).

3.2.4. The effect of pH control on the microbial community structure during pretreatment

Upon the completion of MMP at diverse pH values, we obtained the hydrolysate to analyze the structures of microbial communities. We assigned fragments of 16S rRNA gene at diverse taxa levels, ranging from phylum to genus. The microbial community structure of bacteria at different pH values at the phylum, class, order, and genus levels are illustrated in Fig. 2. MMP at different pH values greatly influenced bacterial community structure. Some strictly anaerobic and uncultured microbes vanished during pretreatment, accompanied by the proliferation of aerobic and facultative microbes.

At the phylum level, *Firmicutes*, *Bacteroidota*, and *Proteobacteria* were dominant during pretreatment, accounting for 86%–99% of the bacterial population. These bacteria are mainly facultative anaerobic bacteria, which are widely active in various anaerobic and microaerobic degradation processes, and play a role in macromolecular hydrolysis and organic acid production (Ziganshin et al., 2013). In the initial mixed raw materials, the *Firmicutes* relative abundance was determined to be 70%. *Firmicutes* have been well-recognized to generate extracellular enzymes (such as protease, cellulase and lipase), and they play important roles in

Table 2
VFA composition of hydrolysate during 5 days of MMP.

Sample	Acetic acid concentration (g/L)	Acetic acid proportion (%)	Propionic acid concentration (g/L)	Propionic acid proportion (%)	Acetic acid/Propionic acid ratio
pH 5	3.37–3.97	81.00–85.76	0.21–0.16	4.71–3.40	17.21–25.22
pH 6	4.21–5.85	77.20–85.88	0.57–0.68	8.68–11.78	6.56–9.89
pH 7	4.76–6.83	80.36–82.45	0.67–1.04	11.39–13.73	7.09–5.98
pH 8	4.10–7.94	79.92–81.82	0.56–1.58	11.09–15.98	7.38–5.01
pH-U	3.47–6.35	78.29–81.10	0.53–0.94	11.59–13.89	7.00–5.84

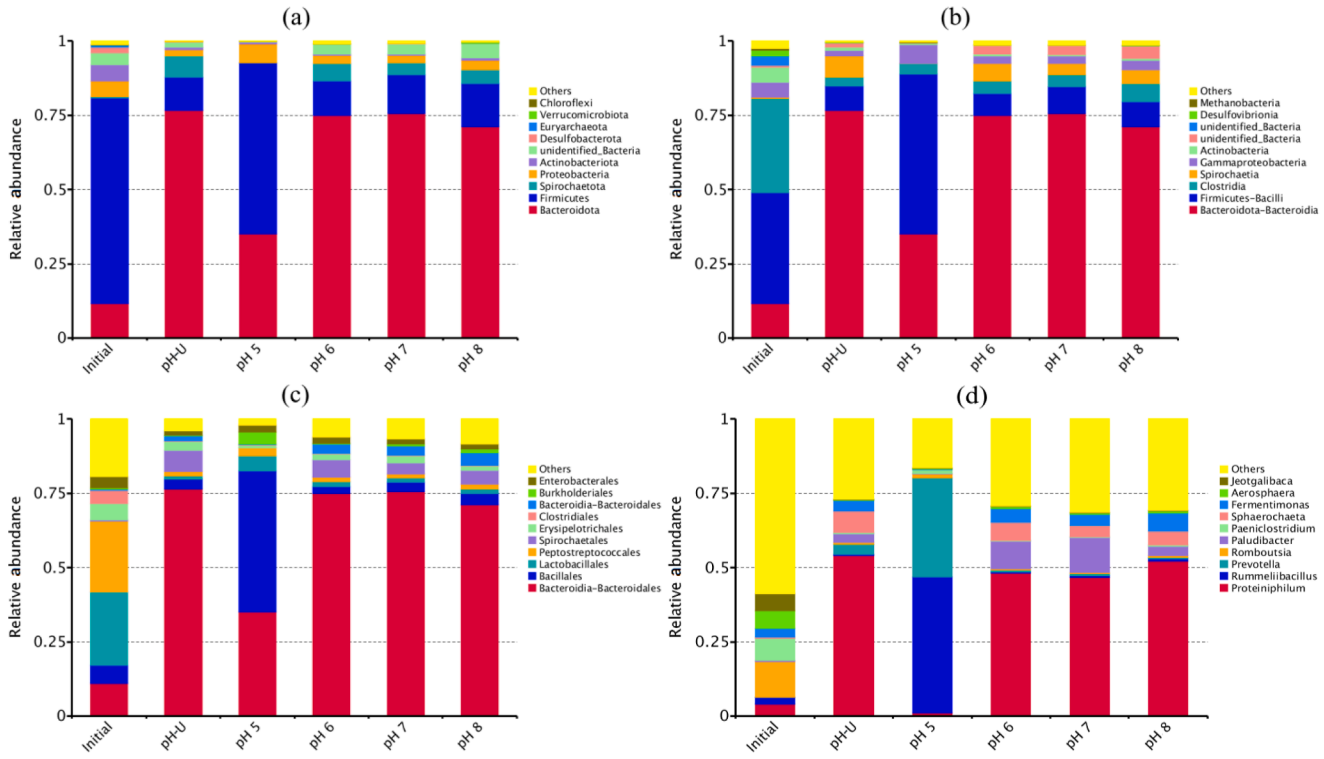


Fig. 2. Microbial community composition of bacterial of pretreatment slurry before and after 5 days MMP at the phylum (a), class (b), order (c), and genus (d) level.

lipid, lignin, protein and cellulose metabolism (Lim et al., 2014). This explains the high extracellular cellulase activity in the initial hydrolysate. After 5 d of MMP, the dominant bacteria in hydrolysate shifted from *Firmicutes* to *Bacteroidota* at pH 6, pH 7, pH 8, and pH-U. The relative abundance of *Firmicutes* reduced to 10%–15% and that of *Bacteroidota* increased to 71%–76%. A metaproteomic study found that *Bacteroidota* is a common bacteria phylum in the decomposition of lignocellulose raw materials, which participates in hydrolyzing polysaccharides such as cellulose and hemicellulose, and suitable for a neutral pH environment (Hanreich et al., 2013). This explains our findings, at pH 6, pH 7, pH 8, and pH-U, Bacteroidota was more suitable for the microaeration pretreatment conditions in this study, and had good hydrolysis ability for lignocellulose. In contrast, at the pH 5 conditions, the *Firmicutes* relative abundance was kept at an increased level relative to those under additional pretreatments, which is due to *Firmicutes* having stronger resistance to low pH than other microbial communities (Pervin et al., 2013).

At the class and order levels, the microbial community structure of bacteria of all pH-controlled MMPs also greatly changed. After 5 d of MMP at pH 6, pH 7, pH 8, and pH-U, strictly anaerobic *Clostridia* and *Lactobacillus* were reduced, and the abundance of *Bacteroidia* increased. At pH 5, *Firmicutes-Bacilli* proliferated due to strong resistance.

At the genus level, unidentified microbial communities accounted for the majority of the bacterial population in the initial mixed raw materials. After 5 d of pretreatment, *Proteiniphilum*, *Rummeliibacillus*, and *Prevotella* were the dominant microbial communities. At pH 6, pH 7, pH 8, and pH-U MMP, *Proteiniphilum* increased rapidly, accounting for 46%–54.0% of the bacterial population. *Proteiniphilum* represents a vital genus of the *Bacteroidota* phylum, and it can promote hemicellulose decomposition and use the N-containing substrates for generating acetic acid and NH_3 (He et al., 2017). At pH 5, *Rummeliibacillus* and *Prevotella* proliferated rapidly, accounting for 46.0 and 33.3% of the bacterial population after 5 d of pretreatment. *Rummeliibacillus* can secrete multiple cellulase and hemicellulase enzymes, which are widely used in the acidification of lignocellulosic material (Weimer et al., 2011). Many articles suggest that *Prevotella* are important acidogenic bacteria, and that low pH is beneficial for *Prevotella* (Lebuhn et al., 2014; Won et al., 2013).

Some studies suggest that increased hydrolysis could possibly be associated with fungi existing within the inoculum, and they can decompose organic biomass in the microaerobic environment (Peces et al., 2016). Fungi can decompose polysaccharides such as cellulose and hemicellulose through producing the extracellular enzymes. In the microaerobic conditions, fungi possibly possess the partial depolymerization complex structures in simple organic molecules; as a result, more corn stover is available for fermentative bacteria. Fig. 3 shows the fungal community structure at different pH levels at phylum and genus scales. At the phylum level, *Ascomycota* (43.4%) was the most abundant, followed by *Basidiomycota*. *Ascomycota* are the most abundant composting phyla in cow manure, food and garden waste, and sewage sludge (Meng et al., 2019; Wang et al., 2018), which produce various enzymes to degrade hemicellulose and cellulose and effectively use the substrate nutrients (Singh et al., 2003). *Basidiomycota* represent the vital contributing factors to the functions of the ecosystem at several levels, which stand for the main lignin degraders (Xing et al., 2020). Initially, at pH 6, pH 7 and pH 8 MMP, *Ascomycota* were represented by the *Trichoderma* genus, which represented the omnipresent cellulosic substance colonizers that are usually detected in the presence of decayed plant materials (Kubicek et al., 2008). Under pH-U and pH 5 conditions, the microbial community structure of fungi at the genus level changed significantly, indicating that pH change greatly impacts fungal community structure.

Ammonia concentration is also an important factor in changing microbial diversity and activity. To evaluate the effect of NH_4-N on MMP, the ammonia concentration under different pH control strategies was measured. As shown in Fig. 1d, the ammonia concentration remained low and should not have an important impact on the MMP process.

In summary, at different pH pretreatment conditions, different microbial community structures were formed, causing changes in the hydrolysis capacity and pathways and leading to different soluble compositions and VFA patterns (Calicioglu et al., 2018).

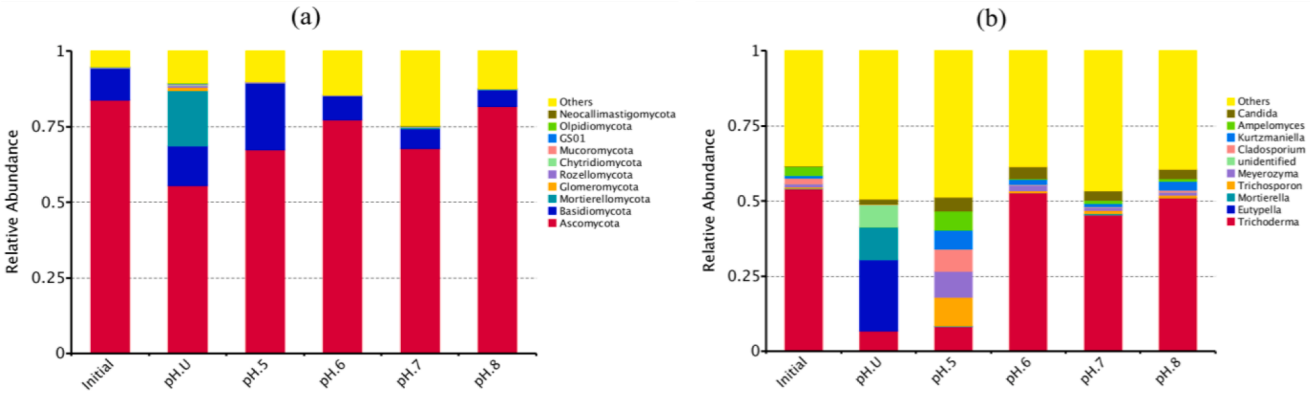


Fig. 3. Microbial community composition of fungal of pretreatment slurry before and after 5 days MMP at the phylum (a) and genus (b) level.

3.3. *Effect of pH and pretreatment time on methane yield*

The methane production during 20 d of mesophilic AD of corn stover under different pH pretreatments for different lengths of time (0, 6, 12, 24, and 48 h) is shown in Fig. 4. The experimental data fitted with Eq. (1) are shown in Table 3.

The methane yield potential was significantly related to pretreatment time and not significantly (p greater than 0.05) related to pH. After 6 h pretreatment, all methane yields during 20 d AD (B_{20}) of pretreated corn stover were higher than that of the no-pretreatment control except for the pH 5 sample; the methane yield sequence of the samples was pH-U > pH 6 > pH 7 > pH 8 > pH 5. The highest cumulative methane yield arrived at 146.5 mL/g TS for the pH-U pretreatment, which was 10.2% higher than that of the control. Nevertheless, the methane yield potential resulting from linear regression had different features. The methane yield potentials of pH-U, pH 5, and pH 6 were essentially equal to the control, whereas the k of pH-U, pH 6, pH 7, and pH 8 were all higher than those of the control, which represented a higher hydrolysis rate of AD through MMP.

After 12 h pretreatment, all pH treatments had greater B_{20} of pretreated corn stover than the 6 h pretreatment counterparts except for pH 5. Both the methane yield potential and hydrolysis rate were enhanced

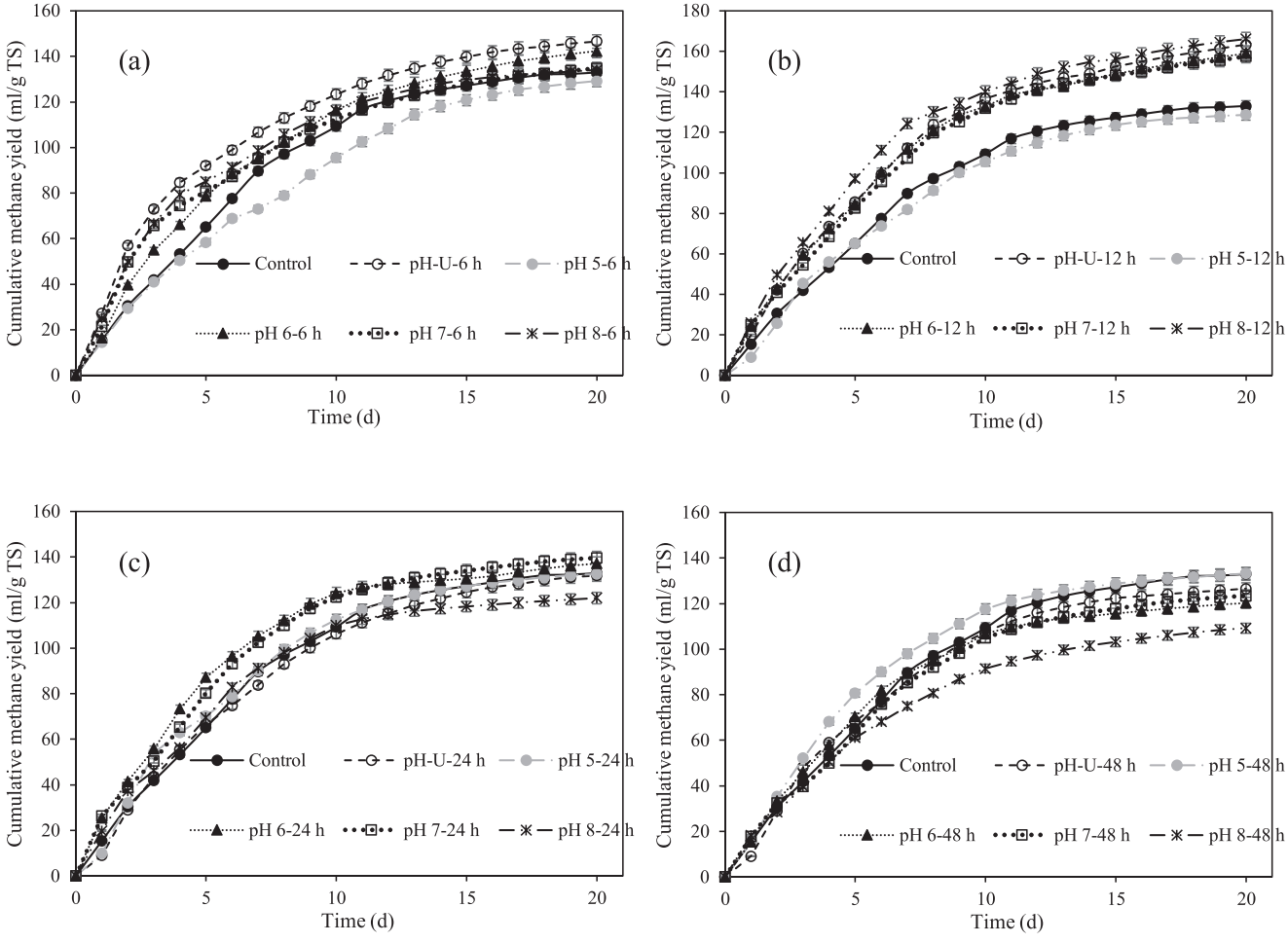


Fig. 4. Cumulative methane yield during AD under different pH pretreatments for 6 (a), 12 (b), 24 (c), and 48 (d) h.

Table 3
Parameters of first-order equation fitting experimental data.

Sample	B_{20}(ml/g $TS_{substrate}$)	B_0(ml/g $TS_{substrate}$)	k (d^{-1})	R^2
Control	132.9 ± 4.5	149.5 ± 5.1	0.125 ± 0.012	0.9971
pH-U-6 h	146.5 ± 4.8	146.4 ± 4.2	0.200 ± 0.011	0.9964
pH 5–6 h	129.2 ± 3.8	154.3 ± 4.9	0.097 ± 0.010	0.9985
pH 6–6 h	142.3 ± 4.0	150.4 ± 5.3	0.146 ± 0.009	0.9996
pH 7–6 h	134.8 ± 4.2	135.8 ± 4.8	0.185 ± 0.008	0.9966
pH 8–6 h	133.9 ± 3.9	135.2 ± 5.6	0.202 ± 0.013	0.9979
pH-U-12 h	163.3 ± 3.6	172.9 ± 6.2	0.146 ± 0.012	0.9988
pH 5–12 h	128.6 ± 4.6	144.9 ± 5.0	0.123 ± 0.011	0.9972
pH 6–12 h	159.0 ± 5.1	166.8 ± 5.3	0.152 ± 0.007	0.9989
pH 7–12 h	157.6 ± 4.9	168.3 ± 6.1	0.144 ± 0.012	0.9984
pH 8–12 h	166.1 ± 5.6	174.3 ± 6.3	0.169 ± 0.011	0.9993
pH-U-24 h	131.8 ± 4.6	147.8 ± 4.2	0.121 ± 0.008	0.9983
pH 5–24 h	155.0 ± 5.0	143.5 ± 5.5	0.143 ± 0.011	0.9971
pH 6–24 h	137.2 ± 4.6	140.3 ± 5.2	0.192 ± 0.013	0.9982
pH 7–24 h	139.7 ± 3.9	147.2 ± 4.8	0.164 ± 0.010	0.9976
pH 8–24 h	122.0 ± 5.3	129.0 ± 4.6	0.168 ± 0.011	0.9967
pH-U-48 h	126.5 ± 4.5	137.8 ± 5.5	0.142 ± 0.007	0.9970
pH 5–48 h	133.2 ± 5.2	140.1 ± 4.0	0.169 ± 0.011	0.9981
pH 6–48 h	120.3 ± 3.6	126.6 ± 5.1	0.167 ± 0.012	0.9979
pH 7–48 h	123.5 ± 3.7	135.2 ± 5.2	0.136 ± 0.010	0.9970
pH 8–48 h	109.2 ± 4.8	115.7 ± 4.4	0.149 ± 0.009	0.9994

compared to the control. The highest B_{20} was 166.1 mL/g TS for the pH 8 pretreatment, which was 25.0% higher than that of the control. For the pH-U pretreatment, the highest methane yield potential was 174.3 mL/g TS, which was 16.6% higher than that of the control. The enhancement was consistent with the results of Fu et al. (2016), who found that the maximum cumulative methane yield was 16.5% higher than that of the control using a microaeration strategy for the AD of corn stover.

After 24 h pretreatment, even when the initial methane production was higher than the control, both the B_{20} and methane yield potential of pretreated corn stover were lower or slightly higher than that of the control. This means that the overconsumption during the pretreatment process negatively affects methane production. The lowest B_{20} was 122.0 mL/g TS for the pH 8 pretreatment, which was 8.2% lower than that of the control.

The methane production results of the 48-h pretreatment verified the speculation detailed in Section 3.2.2, as the negative effect on methane yield potential by over respiration dominated the pretreatment effect. Even when the sCOD was higher than the control and the 24 h pretreatments, the final methane production was lower than the control due to overconsumption during the pretreatment process. The greatest negative effect (12.7%) was detected at 109.2 mL/g TS for the pH 8 pretreatment, which was 17.8% lower than that of the control.

Overall, at pH 5, MMP changed the methane production rate, but not the methane yield potential. Among pH 6, pH 7, pH 8, and pH-U pretreatment, pH had an effect on AD, but the effect was not significant (p greater than 0.05). Pretreatment time significantly affected ($p < 0.05$) methane yield potential, and 12 h pretreatment was the optimal pretreatment time, improving both methane yield potential and methane production rate. Hence, the methane yield results of the optimization study imply that MMP is a simple and cost-effective approach for the effective improvement of anaerobic biological degradability for recalcitrant lignocellulosic biomass. pH and pretreatment time led to both positive and negative AD results. The threshold level of microaeration pretreatment may differ across diverse AD systems, which must be determined for preventing over-pretreatment.

4. Conclusions

MMP resulted in the metabolism of complex lignocellulose substrates by facilitating the proliferation of bacteria and fungi with hydrolysis and acidogenic abilities, which led to a higher AD rate. The proliferation of acidogenic and VFA-utilizing bacteria and aerobic respiration increased substrate reduction, which make against to improving methane yield potential. pH altered hydrolysis and metabolic pathways by affecting microbial diversity. Both pH and pretreatment time affected hydrolysis and methane yield. Maximum methane yield potential was found at 12 h pretreatment with a pH of 8, and was 174.3 mL/g TS and 16.6% higher than that of the control.

CRediT authorship contribution statement

Haipeng Xu: Conceptualization, Methodology, Software, Investigation, Writing - original draft, Writing - review & editing. **Yan Li:** Validation, Formal analysis, Visualization, Software. **Dongliang Hua:** Validation, Formal analysis, Visualization, Project administration. **Yuxiao Zhao:** Resources, Data curation. **Lei Chen:** Resources, Data curation, Writing - review & editing. **Lei Zhou:** Formal analysis, Validation. **Guanyi Chen:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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